How to Present "Customer Shopping Behavior Analysis" — Practical Guide

1. The 30-Second Elevator Pitch (say this first)

> "My project analyzes 3,900 customer transactions from a retail company to understand shopping behavior — who buys what, who spends more, and what drives repeat purchases. I cleaned the raw data using Python, loaded it into a PostgreSQL database, wrote SQL queries to answer 10 real business questions, and built a Power BI dashboard to visualize the findings. The goal was to give the company actionable recommendations to increase revenue and customer loyalty."

Memorize this. Examiners often only need this much to start asking questions — everything else you can expand on when asked.

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2. Explain the Project in Simple Terms (in order)

Walk through it like a story — this order mirrors how you actually did the work, so it's easy to remember.

Step 1: The Business Problem

A retail company wants to know: **"How can we use our customer data to sell more and keep customers coming back?"** They didn't know things like — do discounts actually work? Are subscribers better customers? Which products should we push?

Step 2: The Dataset

- 3,900 customer records, 18 columns (age, gender, product, category, purchase amount, review rating, subscription status, discount, shipping type, previous purchases, payment method, etc.)
- One row = one transaction

Step 3: Data Cleaning (Python/Pandas)

Explain this simply as "getting the data ready before analysis":

- **Missing values**: 37 review ratings were missing → filled with the median rating of that product's category (not overall average — this keeps it fair per category)
- **Column names**: renamed to snake_case (e.g. `Purchase Amount (USD)` → `purchase_amount`) so SQL queries are cleaner
- **New columns created (feature engineering)**:
  - `age_group` — split ages into 4 quartile groups (Young Adult, Adult, Middle-aged, Senior) using `pd.qcut()`

    - Purchase frequency converted from text ("Weekly", "Monthly") into approximate number of days
- **Dropped a redundant column**: `Promo Code Used` was found to be an exact duplicate of `Discount Applied` for all 3,900 rows, so it was removed
- Result: zero missing values, clean dataset loaded into PostgreSQL via SQLAlchemy

Step 4: SQL Analysis (the core of the project)

You answered 10 business questions with SQL. Group them into themes when you explain:

- **Revenue drivers**: gender-wise revenue, age-group revenue
- **Customer value**: who spends above average even with discounts, subscriber vs non-subscriber spend
- **Product performance**: top-rated products, most-discounted products, top 3 products per category
- **Customer segmentation**: New / Returning / Loyal based on previous purchases
- **Loyalty vs subscription**: do repeat buyers actually subscribe?
- **Shipping behavior**: Standard vs Express spend comparison

Step 5: Dashboard (Power BI)

You built an interactive dashboard with:

- KPI cards: 3.9K customers, $59.76 avg purchase amount, 3.75 avg review rating
- Sales/Revenue by Age Group and by Category (bar charts)
- Slicers/filters: Shipping Type, Gender, Subscription Status, Category
- Subscription status donut: 27% Yes / 73% No

Step 6: Key Findings (your headline numbers — memorize these 5)

1. Male customers = 68% of revenue (₹1,57,890 of ₹2,33,081)
2. Clothing dominates: 1,737 transactions (44.5% of total)
3. 80% of customers are "Loyal" (10+ previous purchases) — strong retention
4. Only 27% of customers are subscribed — even among heavy repeat buyers, 72% aren't subscribed
5. Discounts don't suppress spend — 839 discounted transactions still exceeded the average

Step 7: Business Recommendations

Pick your top 3 to speak on confidently:

- Push subscription offers to loyal non-subscribers (biggest untapped opportunity)
- Rationalize discounts on high-discount items (Hat, Sneakers, Coat) since they don't hurt conversion much even at lower depth

- Target female customers more — they're 32% of the base but only 32% of revenue, room to grow

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3. Suggested Presentation Flow (if you get 5-10 minutes)

Time	What to say
0:30	Elevator pitch (Section 1 above)
1:00	Business problem — why this matters to a real company
1:30	Data cleaning in Python — mention the 3 concrete things you did (missing values, renaming, age_group creation)
2:30	Pick 3-4 SQL questions and walk through the logic + result (don't do all 10, examiner will lose interest)
1:30	Show the Power BI dashboard, point at 2-3 visuals
1:00	State 3 key findings and 2 recommendations
0:30	Conclusion — tie back to the business problem

Tip: When showing the dashboard or SQL, point at the actual screen/file rather than describing from memory — examiners respond well to "let me show you" over pure narration.

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4. Likely Cross Questions (and how to answer)

On Data Cleaning

Q: Why did you use median and not mean to fill missing review ratings? A: Median is less affected by outliers/skewed distributions, so it gives a more representative "typical" rating than mean, especially for a bounded scale like 2.5–5.0.

Q: Why per-category median and not overall median? A: Different product categories naturally get different rating patterns; filling with the category's own median keeps the imputation realistic instead of pulling all missing values toward one global number.

Q: Why did you drop the Promo Code column? A: It was 100% identical to Discount Applied for every row — keeping both would be redundant and could mislead correlation analysis.

Q: What is `pd.qcut()` and why use it for age groups? A: `qcut()` splits data into quantile-based bins so each group has roughly equal numbers of customers, rather than `cut()`

which uses fixed-width ranges. This gives fairer demographic comparisons.

On SQL

Q: Explain the difference between WHERE and HAVING — did you use either? A: WHERE filters rows before grouping; HAVING filters after aggregation. In this project, most filtering (e.g. `discount_applied = 'Yes'`) happens in WHERE since it's row-level, not on aggregated results.

Q: Explain the subquery in Q2. A: `SELECT AVG(purchase_amount) FROM customer` runs first to compute the average, then the outer query compares each customer's purchase_amount against that value — this is a scalar subquery.

Q: What is ROW_NUMBER() and why did you use it in Q8? A: It assigns a rank to rows within a partition (here, `PARTITION BY category`) ordered by order count — letting you pull the "top N per group" instead of overall top N.

Q: Why PostgreSQL and not MySQL/SQLite? A: PostgreSQL supports advanced SQL features used here like window functions (`ROW_NUMBER() OVER`), strong type casting (`::numeric`), and integrates well with Python via SQLAlemy/psycopg2 — good fit for analytical queries.

On the Dataset/Findings

Q: Is 3,900 records a large enough sample to draw business conclusions? A: It's a reasonable sample for exploratory analysis and trend-spotting, but for high-stakes business decisions you'd want to validate findings against a larger, more recent dataset or run statistical significance tests.

Q: Your findings say discounts don't reduce spend — could that be misleading? A: Fair point — this is a correlation, not proven causation. Customers who like a product might buy it regardless of discount; the discount didn't necessarily cause the higher spend. I noted this as an insight, not a guaranteed causal relationship.

Q: Why do only 27% subscribe if loyalty is so high (80%)? A: That's actually one of the more interesting findings — it suggests subscription is currently not perceived as valuable enough by customers, even ones who buy often. That's exactly why I recommended targeted subscription offers for that segment.

On Tools/Process

Q: Why Python for cleaning instead of doing it directly in SQL? A: Python/Pandas is better suited for iterative cleaning tasks like handling missing values, feature engineering, and column renaming before the data is structured — SQL is better once the data is already clean and you're doing aggregation/analysis.

Q: What would you do differently with more time? A: Good answer to have ready — e.g. "add time-series analysis if purchase dates were available," "run statistical tests to validate the discount/spend relationship," "build a predictive model for churn or next purchase."

5. Future Scope (have 4-5 ready, don't just list all — pick relevant ones live)

1. **Predictive modeling**: Build a churn prediction model to flag loyal customers likely to stop purchasing, and a "propensity to subscribe" model.
2. **Time-series / seasonality analysis**: If purchase timestamps were available, analyze sales trends over months/seasons instead of only the static `Season` column.
3. **Customer Lifetime Value (CLV)**: Calculate CLV per segment to prioritize marketing spend on the most valuable customers.
4. **A/B testing framework**: Test whether reduced discount depth on high-discount products (Hat, Sneakers, Coat) actually maintains conversion, as recommended.
5. **Real-time dashboard**: Connect Power BI to a live database instead of a static snapshot for ongoing monitoring.
6. **Recommendation engine**: Use purchase history and category affinity to suggest cross-sell products (e.g. pairing top-rated items with high-volume items as already noted in recommendations).
7. **Expand data**: Bring in marketing spend, customer support tickets, or return/refund data for a fuller picture of the customer journey.

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6. Quick Confidence Boosters

- If you don't know an answer, it's fine to say: *"That's something I'd explore in a future iteration of this project — I focused on the descriptive analysis using the available fields."* This is honest and turns a gap into a "future scope" answer.
- Know these 3 numbers cold: **3,900 records**, **80% loyal customers**, **27% subscribed**. These come up in almost every follow-up.
- Practice explaining ONE SQL query end-to-end (pick Q7, the segmentation one — it's the most "story-like" and uses CASE WHEN, which examiners like to ask about).